



Title	Use of University of Nottingham Gas Safety Calculator
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Purpose

This document provides a series of example calculations to aid the use of the in-house University of Nottingham Gas Safety calculators.

The calculators are hosted on the University website

(<https://uniofnottm.sharepoint.com/sites/UoNHealthSafety/SitePages/Compressed-Gas-&-Cryogen-Calculator.aspx>)

The outcome of the calculation can be used as a screening tool to estimate risks to persons in the event of catastrophic and slow leaks of a variety of compressed gases, cryogenic materials, and dry ice.

It is noted that these calculations only form part of the risk assessment, and evidence that a calculation is carried out is not by itself considered a suitable and sufficient assessment of risk. The risk assessment process should still be carried out as normal, and the calculations should only be used to determine who can be harmed, how, and identify the necessary controls.

Note:

This calculator provides estimates for **informational purposes only**. Assessing exposure to gas is a highly complex field - real-world conditions including gas stratification, turbulent mixing, room geometry, temperature, and ventilation dynamics can differ significantly from the idealised models used here.

This tool must not be used as a replacement for a full risk assessment – it is designed to help you form an opinion but cannot take into account all the possible variables required to accurately provide a recommendation.

If there is any doubt about the risk to people, **specialist advice must always be sought** from a competent person, or your health and safety coordinator.

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Outline of Calculator Functionality

This calculator estimates the concentration of a gas released into a room or the fraction of a gas of interest that remains in the air after a hazardous release from compressed gas cylinders, liquefied gas containers, cryogenic vessels, or dry ice.

It also models the localised concentration within a Personal Bubble (12 m³) around the leak source, providing a proximity-risk assessment for anyone working near the gas.

It supports six input scenarios:

- **Known Volume:** You already know the volume of gas (m³) in the cylinder and want to calculate the resulting concentration so you can determine the room concentrations in a catastrophic loss of the cylinder's contents.
- **Compressed Gas:** Calculates the gas volume from cylinder pressure and water capacity of the cylinder. The output of this can then be used to determine the room concentration of the gas in the event of a catastrophic release.
- **Liquefied Gas:** Calculates the gas volume from the weight of the contents and the specific volume of a known gas in the cylinder. The gas volume can then be used to determine the room concentration of the gas in the event of a catastrophic release.
- **Cryogenic Vessel:** Calculates gas volume from the liquid volume using expansion ratios for Liquid Nitrogen, Liquid Helium or Liquid Carbon Dioxide. This can then be used to determine the room concentration of the gas in the event of a catastrophic release.
- **Dry Ice (Carbon Dioxide (CO₂)):** Calculates CO₂ gas volume from a provided dry ice mass. This can then be used to determine the room concentration of the gas in the event of a catastrophic release.
- **Slow Leak:** Estimates the steady-state concentration from a continuous gas leak in a ventilated room, alongside a "Time to Hazard" calculation that shows how long it will take for conditions to become dangerous if ventilation fails.

Assumptions:

The calculator relies heavily on idealised scenarios to provide a functional first-pass screening estimate for basic risk assessment purposes. Some assumptions are conservative, but the model should not be described as uniformly conservative across all gases and release conditions. The assumptions made are as follows:

Perfect Mixing (Uniform Dispersion): The core assumption of almost all basic gas calculators. It assumes that upon release, the gas instantly and perfectly mixes with the air in the specified volume (either the whole room or the Personal Bubble), creating a homogenous atmosphere.

Ideal Gas Behaviour: Calculations for compressed gas volume ignore the compressibility factor (Z-factor) of real gases. While Boyle's Law provides a close and computationally simple estimate, real gases deviate from ideal behaviour at very high pressures. This applies particularly to both polar molecules and larger molecules with high polarisability (where volume will be overestimated) and very light molecules (H₂, He) where volume may be underestimated.

Complete Displacement Mechanics: The calculations assume the released gas forces an equivalent volume of perfectly mixed 'room air' out of the space, rather than either displacing based on gaseous properties or creating a highly pressurised vessel.

Isothermal Conditions: The calculations assume the gas release does not drastically drop the room temperature. In reality, large releases (especially of cryogenics or rapidly expanding compressed gases) will significantly lower the room temperature due to the Joule-Thomson effect.

Room Free Volume: The calculator offers a 40% reduction toggle for cluttered rooms. This is an arbitrary blanket estimate of the physical space occupied by benches, analytical equipment, and cabinets, used when precise free-volume measurements are unavailable.

Preparing for Calculations:

Ahead of performing gas safety calculations, a range of information is needed up front:

1. **The properties of the gas(es)** - Some gases pose multiple threats, (Asphyxiation, Toxic and/or Flammable) so each of the potential hazards need to be considered in isolation and the hazards that present the highest risk should be considered first when identifying mitigations. For example, CO₂ can create an oxygen deficient environment, but it is also toxic. Both hazards need to be evaluated separately, and the controls identified need to address both hazards if necessary. Similarly, for CO, this presents a risk of asphyxiation, toxicity, as well as a flammability hazard – the toxicity is, once again, the highest risk to users.
2. **Nature of the gas in the cylinder:** Where the volume of gas in the cylinder is unknown, the calculation approach differs for liquefied and compressed gases due to their different physical behaviours. As a result, different input parameters are required. The values that may be needed for each gas type are listed below.

Calculation Type	Required Information
Compressed Gas	• Pressure (bar); • Water capacity of cylinder (the volume of water the cylinder is capable of holding in Litres)
Liquefied Gas	• Weight of cylinder contents (NOT including cylinder itself) (kg) • Specific volume (m³/kg) of gas in cylinder
Cryogenic Vessel	• Liquid volume (Litres) • Gas type (LN₂, LHe, LCO₂)
Dry Ice (CO₂)	• Dry ice mass (kg)
Slow Leak	• Leak rate (m³/hr) • Air changes per hour (ACH)

For all calculations, the free air volume of the room is required – this is the available volume within a room that a gas could expand into. It can be calculated by subtracting the size of all solid objects within the room (e.g. furniture and equipment) from the volume of the room itself.

Approach to Gas safety calculations:

The gas calculator works in the following way:

1. Determination of the volume of gas that could be released into a room upon full release of contents of the cylinder.
2. Comparison of released gas volume with the free air volume of the room.

This returns either i) the composition of the contaminant in air or ii) the concentration of oxygen in the air after dilution.

3. Determination of the percentage of gas in the room and comparison against workplace exposure limits, lower explosive limits, or other relevant limits to determine risk to persons in the area.
4. Recommendation for detection provided based upon whether limits are exceeded.

The rest of this document will provide some examples of scenarios for you to work through, and both short answers and full solutions are then outlined for each scenario.

Part 1: Test Scenarios

Scenario 1: Compressed Gas (Asphyxiant)

You have a compressed nitrogen cylinder (asphyxiant) with an initial pressure of 230 bar and a cylinder water capacity of 50L.

The room you are storing it in has a free air volume of 100m³.

1. What is the total volume of gas contained in the cylinder?
2. What would the resulting room O₂ concentration be in the event of a total release?
3. What would the O₂ concentration be within a person's immediate "Personal Bubble" (a 12 m³ zone)?

Scenario 2: Liquefied Gas (Toxic)

You are working with a 10 kg cylinder of liquefied Ammonia gas (weight of contents only). The specific volume of Ammonia is 1.4m³/kg. The room volume is 50m³, but you have a lot of equipment and benches and don't know the exact free air volume.

1. What is the calculated volume of Ammonia gas in the cylinder?
2. What is the final room concentration of Ammonia following release in parts per million (ppm)?

Scenario 3: Cryogenic Vessel (Asphyxiant)

You have a dewar containing 25 Litres of Liquid Nitrogen (LN₂). If fully released and vaporised, the nitrogen will expand into the surrounding space. The room has a free air volume of 150m³.

1. How many cubic metres of gas would be generated following a full release of the contents of the dewar?
2. According to the calculator's thresholds, what is the resulting room O₂ concentration, and what specific recommendation or warning does the calculator provide for this level?

Scenario 4: Known Volume (Carbon Dioxide (CO₂) Enrichment)

You are venting a known volume of 1.5m³ of Carbon Dioxide (CO₂) into a room with a free air volume of 40m³.

1. What is the room concentration of CO₂ as a percentage?
2. Based on the calculator's thresholds, is fixed CO₂ detection required?

Scenario 5: Slow Leak (Flammable)

A faulty regulator causes a measured, continuous leak of 1m³ per hour of flammable gas. The free air volume of the room is 80m³ and the ventilation system provides 6 Air Changes per Hour (ACH).

1. What will the steady-state concentration of the gas in the room be as a percentage?

Scenario 6: Toxic Gas, External Lookups & Detection Placement

You are planning to store a standard BOC size 'AV' cylinder of compressed Carbon Monoxide (CO) at an initial pressure of 51 bar. Carbon monoxide is a toxic gas and chemical asphyxiant with a secondary flammability hazard. The cylinder is located in a room with a free air volume of 80m³.

1. Look up the standard BOC size 'AV' cylinder, and indicate the volume of the contents in the cylinder.
2. Assuming a full release of the contents of the cylinder, what is the resulting room concentration, expressed in parts per million (ppm)?
3. Look up the standard Workplace Exposure Limit (WEL) for Carbon Monoxide in the HSE's EH40 document (or reference the calculator's internal guidance). Compare your calculated CO concentration against the WEL. Does this comparison indicate that detection is required?
4. According to the calculator's methodology report, what specific type of detector should you use for this gas, and at what height should it be mounted?

Scenario 7: Slow Leak, Time to Hazard & Heavy Gas Detection

A regulator on a Carbon Dioxide (CO₂) system has failed. The regulator has an outlet pressure of 2.5 bar, and the leak is equivalent to gas escaping through a pin-hole sized opening about 0.2mm across (equivalent hole diameter).

The room is a small preparation room with 20m³ of free air volume, and the room currently has active ventilation providing 2 Air Changes per Hour (ACH).

1. Using the calculator's leak rate estimator, what is the steady-state CO₂ concentration in the room as a percentage?
2. Suddenly, the building's ventilation system completely fails (0 ACH worst-case scenario). Using the calculator's "Time to Hazard" formula for CO₂, how long will it take for the room to reach the 0.5% detection threshold?
3. According to the calculator's guidance, what specific type of sensor must be used for CO₂, and how high off the floor should it be mounted?

Scenario 8: Risk of Exposure when handling Dry Ice

A worker is handling a container of dry ice (solid carbon dioxide (CO₂)) on the benchtop in order to create a cooling bath. They have 200g of dry ice, and the free air volume in the lab is 50m³.

1. What is the maximum volume of CO₂ gas that could be generated by the complete sublimation of all the dry ice?
2. Determine if a local exposure risk is present to the user when handling the dry ice, and whether further control is needed.

Scenario 9: Mixed Gas Cylinders & Multiple Independent Risks

You are using a standard BOC size 'L' cylinder containing a mixed gas comprising 10% Carbon Dioxide, 10% Hydrogen, and 80% Nitrogen at an initial pressure of 200 bar. The cylinder is located in a small instrument room with a 30m³ free air volume.

1. Visit the BOC website to locate the details of the cylinder.
From the data on the website, determine the volume of gas in the cylinder.
2. When assessing mixed gases in the calculator, you must assess each component gas independently. First, calculate the total volume of gas in the cylinder.
3. Next, calculate the individual gas volumes for the CO₂ (10% of total), Hydrogen (10% of total), and Nitrogen (80%) and determine their specific room concentrations.
4. Finally, calculate the total Oxygen depletion caused by the entire volume of gas displacing the room air.
5. Based on the concentrations calculated in 3 and 4, which warning threshold is likely to be breached first in the event of a leak? Therefore, what specific detector type is the most critical to install, and are the others redundant?

Scenario 10: Dense Gas Stratification

A worker is in a plant room. The room has two tiers, separated by a metallic grid mesh floor. The lower tier forms a confined basement area with a solid floor area of 15m². The upper area has the same floor area. The ceiling height (taking into account the height across both tiers) is 3m

An Argon cylinder fails, releasing the full contents of the cylinder (3m³) into the room. The gas is significantly heavier than air, so it falls through the gaps in the grid mesh floor and pools at the bottom of the lower tier.

The height of the stratified layer (assuming no passive diffusion/poor ventilation, etc) can be expressed as $H = V/A$, where V is the volume of gas in m³ and A is the floor area in m².

1. Using the formula above, calculate the height of the stratified gas cloud after release.

2. Calculate the volume of the room and assume a 40% reduction is needed to determine the free air volume. (Assume the 40% reduction is required across the whole volume of the room).
3. Determine the O₂ concentration for the following situations:
 - a. If the gas perfectly stratifies at the height calculated in 1.
 - b. The room concentration after equilibration of the Argon gas within the whole room.
4. Assuming the worker is in the lower part of the room, (and they begin working in the lower area after the gas has perfectly stratified) and they are working at a height of 0.5m, would it appear that there is any risk to them? Is this strictly true, and how might this risk evolve over time?

Part 2: Answers & Methodology

Scenario 1: Compressed Gas (Asphyxiant)

Answers:

- **Gas Volume:** 11.35m³.
- **Room O₂ Concentration:** 18.57%.
- **Personal Bubble O₂ Concentration:** 1.13%.
 - *Methodology:* The personal bubble represents a fixed 12m³ volume immediately surrounding the release point. As 11.35m³ of gas replaces most of the air in this space, the local oxygen level is significantly reduced.

Explanation:

In this scenario the cylinder pressure and water capacity are already known. This information is usually provided by the gas supplier. In some cases, the gas volume is stated directly, meaning that this calculation step can be skipped.

In this instance, we can select: “I have a **compressed** gas cylinder, but do not know the volume of the contents.”, enter the values and click “Calculate Volume” – shown in the screenshots below.

Calculation Type
 Select the scenario that matches your situation

Known Vol

Compressed

Liquefied

Cryogenic

Slow Leak

Choose an option based on what information you have available about your gas/cryogen source.

☐

 I have a gas cylinder and **already know the volume** of the contents.

☒

 I have a **compressed** gas cylinder, but do not know the volume of the contents.

Calculate Volume – Compressed Gas
 From cylinder pressure and water capacity

Initial pressure (bar)

Cylinder water capacity (L)

Volume of gas in cylinder = **11.35 m³**

✓ Your calculated value will be pre-populated in Step 4 below.

You need the **initial pressure** (from the cylinder gauge, in bar) and the **water capacity for the cylinder** (which can usually be located on the datasheet for the cylinder, in litres).

After clicking Calculate Volume, we obtain a value of 11.35m³, and this corresponds to the volume of gas within the cylinder.

After this, we continue through the calculator and are prompted to enter the room's free air volume. The gas volume calculated in the previous step is automatically populated in the relevant field, so no manual entry is required.

We input the free air volume and selected “Asphyxiant” as the gas hazard type.



Calculate Concentration in Room

Enter room volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)

Volume of gas (m³)

Gas Hazard Type

Asphyxiant (O₂ Depletion) ▼

 **Calculate Concentration**

 **Generate Report**



The **free air volume** is the room volume minus the space occupied by furniture and equipment. If unknown, apply the 40% reduction checkbox as a conservative estimate.

After this, the concentration of O₂ in the room is displayed in the event of a catastrophic release.

O₂ concentration in room: **18.57%**

 Further assessment of risk necessary. Your assessment should account for local conditions including airflow, potential gas build-up, and how the specific asphyxiant behaves in the space.

Similarly, a personal bubble calculation is automatically calculated as part of the process. This simulates the O₂ concentration in a 2x2x3 bubble immediately around the cylinder.

O₂ concentration near leak: **1.13%**

 Immediately dangerous to life for anyone near this leak. Oxygen levels in the personal zone would drop to a level that causes loss of consciousness within seconds and can be fatal. No person should approach, work near, or pass through this area without self-contained breathing apparatus (SCBA) and a rescue plan in place.

Scenario 2: Liquefied Gas (Toxic)

Answers:

- **Gas Volume:** 14m³.
- **Room Concentration:** 466,666.67ppm.

- *Methodology:* The 40% reduction drops the 50m³ room to an effective free air volume of 30m³.

Explanation:

This scenario is asking us to determine the volume of Ammonia gas that can be contained within a cylinder of liquefied contents. The phase of the medium is different compared to a compressed gas cylinder, but we can still calculate the volume of gaseous contents that the cylinder contains.

This time, we select: “I have a liquefied gas cylinder, but do not know the volume of the contents.”

This scenario provides the mass of the gas contents only. Care should be taken, as some suppliers quote the combined weight of the cylinder and its contents; for this calculation, only the mass of the gas is required. The specific volume of gas is also provided. This value is gas-specific and can usually be found from reliable online sources. Engineering Toolbox is a useful reference, although not all gases are listed there.

We simply enter both the weight of contents and specific volume and click “Calculate Volume”.

Calculate Volume - Liquefied Gas
From weight and specific volume

Weight of cylinder contents (kg)
10

Specific Volume (m³/kg)
1.4

Calculate Volume

Volume of gas in cylinder = **14.00 m³**

✓ Your calculated value will be pre-populated in Step 4 below.

You need to know the weight of the **contents only** (subtract the weight of the cylinder itself). Find the **specific volume** from the gas datasheet or [Engineering Toolbox](#).

This calculation gives a volume of 14m³.

As before, we move further down the calculator to determine the final room gas concentration following a catastrophic release.

The room volume is 50m³, but as the amount of furniture and equipment is unknown, a 40% reduction is applied to estimate the free air volume. This is a conservative approach; where rooms are known to be heavily cluttered, a more accurate free air volume can be obtained by subtracting the volume of equipment from the total room volume.

For this scenario, "Toxic" is selected as the gas hazard type before running the calculation.



Calculate Concentration in Room

Enter room volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

☒ Apply 40% reduction for conservative estimate (if free air volume unknown)

Volume of gas (m³)

Gas Hazard Type

 **Calculate Concentration**

 **Generate Report**



The **free air volume** is the room volume minus the space occupied by furniture and equipment. If unknown, apply the 40% reduction checkbox as a conservative estimate.

For toxic gases, the calculated concentration on its own is not enough to make a recommendation, as exposure limits vary between chemicals. The result must be compared with the appropriate workplace exposure limit (WEL).

In this example, the calculated room concentration of 466,667 ppm would far exceed typical WELs, meaning additional controls would likely be needed for most toxic gases to protect users.

For completion, the WEL for ammonia is 25ppm, so a catastrophic loss would greatly exceed the exposure limit, and controls would be a necessity.

Gas concentration: **466666.67 ppm**

Please compare this to the [WEL for your gas](#) (EH40). Detection is needed if the calculated value exceeds the WEL.

Common WELs: Ammonia 25 ppm, Carbon Monoxide 30 ppm, Hydrogen Sulphide 5 ppm.

Your risk assessment should also consider local environmental influences, including how the toxic gas disperses, ventilation patterns, and any confined spaces where exposure concentrations could rise.

Consider the location of the cylinder, and how closely a person must work in relation to it. There may be instances where the calculator may not suggest detection is necessary due to the room concentration being below a threshold, however exposure limits for toxic gases are generally low enough such that a user can exceed the WELs within seconds after exposure even to small leaks.

Scenario 3: Cryogenic Vessel (Asphyxiant)

Answers:

- **Gas Volume Generated:** 17.35m³.
 - *Methodology:* Liquid Nitrogen (LN₂) expands dramatically, multiplying by an expansion ratio of approximately 694.
- **Room O₂ Concentration & Recommendation:** 18.53%.
 - *Recommendation:* The calculator categorises levels between 18% and 19.5% as requiring further risk assessment, noting to account for local conditions, airflow, and gas build-up.

Explanation:

When cryogenic liquids boil, they rapidly expand creating hundreds of times the volumes of the initial liquid. This can very quickly create an asphyxiation risk.

We begin by calculating the volume of the gas after expansion by selecting the option: “I have a **Cryogenic** vessel (Dewar / Tank).”



Calculation Type
Select the scenario that matches your situation


Known Vol


Compressed


Liquefied


Cryogenic


Slow Leak

Choose an option based on what information you have available about your gas/cryogen source.

☐

 I have a gas cylinder and **already know the volume** of the contents.

☐

 I have a **compressed** gas cylinder, but do not know the volume of the contents.

☐

 I have a **liquefied** gas cylinder, but do not know the volume of the contents.

☒

 I have a **Cryogenic** vessel (Dewar / Tank).

Next, select the cryogen being used and enter the known contents of the vessel. If the contents are uncertain, use the maximum supported capacity (the largest volume the vessel is designed to safely contain).

In this instance, we know it holds 25L of liquid nitrogen (LN₂), so we enter that into the calculator and click “Calculate Volume”.

Calculate Volume – Cryogenic Vessel

Liquid nitrogen or liquid helium

Cryogen Type

Liquid Nitrogen (LN₂)

Liquid Volume (Litres)

25

Calculate Volume

Volume of gas generated = **17.350 m³**

This shows us that this dewar could release up to 17.35 m³ of nitrogen gas.

Scrolling down, we can then calculate the room concentration. The free air volume is already known and is automatically populated in the calculator so not manual entry is required. We then click "Calculate Concentration"

O₂ concentration in room: **18.53%**

Further assessment of risk necessary. Your assessment should account for local conditions including airflow, potential gas build-up, and how the specific asphyxiant behaves in the space.

The calculator indicates that further risk assessment is required.

Although room ventilation is likely to prevent oxygen levels from dropping to hazardous levels, there may still be a local asphyxiation risk if a user works close to the source of the leak. When liquid nitrogen is released, the cold nitrogen gas produced may initially collect at low level before warming and mixing, increasing the risk of localised oxygen depletion near the floor.

In situations like this, the hierarchy of control should be applied. Where possible, risks should be reduced by measures such as using smaller dewars (where liquid nitrogen is essential) and ensuring that ventilation is sufficient to control residual risk. Gas detection can also be considered to alert users to leaks, but whether this is required should be determined through risk assessment, taking into account working practices, occupancy and existing controls.

In addition to the gas risk, the assessment must also consider the hazards associated with direct exposure to cryogenic liquids.

Scenario 4: Known Volume (CO₂ Enrichment)

Answers:

- **Room Concentration:** 3.75%.
- **Detection Required?:** Yes.

- **Methodology:** The calculator states that CO₂ detection is required if the concentration is above 0.5% (or 5000ppm) (these values are the WELs for CO₂ as found in EH40)

Explanation:

In this example, we are provided with the volume of gas in the cylinder already (1.5m³). Some suppliers will provide this information up front on their website or guidance sheet. Knowing this makes the calculation much easier to complete, and also more accurate the gas contents is usually measured rather than calculated.

Knowing the volume of CO₂ in the cylinder already, we can select the option: “I have a gas cylinder and already know the volume of the contents.”

From there, we simply enter the values from the scenario, select Carbon Dioxide from the Gas Hazard Type dropdown, and click “Calculate Concentration”.

Calculation Type

Select the scenario that matches your situation

Known Vol

Compressed

Liquefied

Cryogenic

Slow Leak

Choose an option based on what information you have available about your gas/cryogen source.

☒ I have a gas cylinder and **already know the volume** of the contents.

Calculate Concentration in Room

Enter room volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

40

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)

Volume of gas (m³)

1.5

The **free air volume** is the room volume minus the space occupied by furniture and equipment. If unknown, apply the 40% reduction checkbox as a conservative estimate.

Gas Hazard Type

Carbon Dioxide (Enrichment)

Calculate Concentration

Generate Report

This displays a room concentration of 3.75% CO₂. As indicated below, the recommendation is such that detection must be considered as this level of gas release is well above the 0.5% limit.

Gas concentration in room: **3.75%**

● CO₂ detection required. Your risk assessment should also consider local conditions such as low-level accumulation, ventilation effectiveness, and whether CO₂ may collect in pits, trenches, or sunken areas.

Scenario 5: Slow Leak (Flammable)

Answer:

- **Steady-State Concentration: 0.21%.**

Explanation:

Calculating the volume of a slow leak is not an exact science, and this calculator should be used with caution. If values are not known or measurable, then coming to an accurate conclusion on the risk to a person is difficult. Where possible, avoid estimating leak rates and air-change rates; these should be measured rather than estimated.

In this scenario, the leak rate is measured as 1 m³/hr. We begin by selecting the slow leak calculator:

The screenshot shows the 'Calculation Type' section of the calculator. It includes a title 'Calculation Type' with a bell icon and the instruction 'Select the scenario that matches your situation'. Below this is a row of five icons representing different gas sources: 'Known Vol' (a box), 'Compressed' (a gas cylinder), 'Liquefied' (a gas cylinder with a liquid level), 'Cryogenic' (a Dewar flask), and 'Slow Leak' (a clock icon). To the right of these icons is the text 'Choose an option based on what information you have available about your gas/cryogen source.' Below this row are five radio button options. The first four options are for gas cylinders: 'I have a gas cylinder and **already know the volume** of the contents.', 'I have a **compressed** gas cylinder, but do not know the volume of the contents.', 'I have a **liquefied** gas cylinder, but do not know the volume of the contents.', and 'I have a **Cryogenic** vessel (Dewar / Tank)'. The fifth option, which is selected with a blue dot, is 'I want to calculate for a **Slow Leak** / Constant Release with Ventilation.'

After this, we can enter all the details from the scenario into the calculator. We enter the measured leak rate of 1 m³/hr, then the air changes per hour (6 ACH) and the free air volume (80 m³). We then select Flammable as the Gas Hazard Type and click Calculate.

Enter the Room Volume, Leak Rate and ACH, then click Calculate.

Free Air Volume of room (m³)

80

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)

Slow Leak Parameters

Leak Rate (m³/hr) ⓘ
Small ≈ 0.001 | Medium ≈ 0.1 | Large ≈ 1 | Catastrophic ≈ 100

1

Air Changes per Hour (ACH)

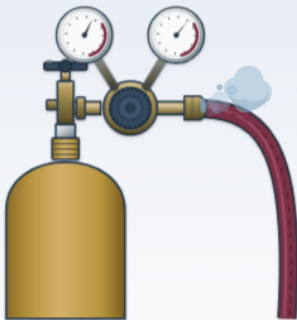
6

Note: This calculation assumes perfect mixing of the gas within the room.

Gas Hazard Type

Flammable

The free air volume is the room volume minus the space occupied by furniture and equipment. If unknown, apply the 40% reduction checkbox as a conservative estimate.

For a **slow leak** (e.g., from a regulator or valve), gas mixes continuously with room air. Enter the **leak rate** and **ACH** to find the steady-state concentration.

 **Calculate Concentration**

 **Generate Report**

We end with a steady-state concentration of 0.21%. This is below 25% of the value of LELs for common flammable gases such as hydrogen and methane. The need for detection remains dependent on the specific gas, its LEL, local accumulation, ignition sources, and the proximity of work.

Gas concentration in room: **0.21%**

■ Please compare this to the [LEL for your gas](#) to confirm whether detection is required.
Detection is required if the calculated percentage exceeds 25% of the LEL.

Common LELs: Acetylene 2%, Hydrogen 4%, Methane 5%.

Your risk assessment should also consider local conditions relevant to your specific flammable gas, such as whether it tends to rise or sink, how ventilation affects dispersion, and the location of potential ignition sources.

⚠ **Worst-Case Time to Hazard (if ventilation fails)**

🕒 Gas accumulates at **1.250% per hour** in a sealed room. Compare to 25% of the LEL for your gas.

Based on 0 ACH (sealed room). Assumes all gas accumulates with no air exchange.

Scenario 6: Toxic Gas, External Lookups & Detection Placement

Answer:

- **Gas Volume:** 0.5m^3 .
 - Methodology: A standard BOC size 'AV' cylinder has a cylinder capacity of 500 Litres. To convert from L to m^3 , we divide by 1000, which equals 0.5m^3 of CO.
- **Room Concentration:** 6250 ppm.
- **WEL & Detection Requirement:** Yes, detection is required. The internal guidance notes a common WEL for CO is 20 ppm.
- **Detector Type & Placement:** Electrochemical sensor, mounted at breathing zone height (approx. 1.5m - 1.8m).
 - Methodology: Toxic gases require electrochemical sensors. CO has a density similar to air, so it is mounted at typical breathing height.

Explanation:

When calculating risks from compressed gas or cryogen hazards, we will likely need to search for the relevant data in order to carry out calculations. Different providers will provide different information which can make the process harder.

In this instance, we know the gas is Carbon Monoxide and the cylinder is an AV cylinder from BOC. We will need to visit their website to locate the details for the cylinder.

In this instance, the site lists 500L as the capacity for an AV cylinder, which corresponds to 0.5m^3 of gas.

Carbon Monoxide N3.7 (99.97%), Research Grade, Cylinder, 500L

Part Number: 270236-AV

Supply Option

Available in cylinders, contact us for alternative sizes

Size

[Size guide](#) →

AZ - 20L

AV - 500L

AZ - 100L

As we already know the gas volume (500L = 0.5m³), we can skip that calculation and go straight to “I have a gas cylinder and already know the volume of the contents.”



Calculation Type
Select the scenario that matches your situation

 Known Vol

 Compressed

 Liquefied

 Cryogenic

 Slow Leak

Choose an option based on what information you have available about your gas/cryogen source.

☒

 I have a gas cylinder and **already know the volume** of the contents.

After selecting that, we enter the gas volume and free air space, as normal. We then select Toxic under the Gas Hazard Type and then Calculate Concentration.

By clicking calculate, we see that the room concentration will reach 6250 ppm if there is a catastrophic release.



Calculate Concentration in Room

Enter free air volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

80

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)

☐ Dense Gas (advanced) 

Volume of gas (m³)

0.5



The **free air volume** is the room volume minus the space occupied by furniture and equipment. The most accurate way to obtain this value, is to directly measure it. There are some phone apps that may help you do this. If unknown, enter the room volume and apply the 40% reduction checkbox as a conservative estimate.

Gas Hazard Type

Toxic

Gas concentration: **6250 ppm**

 Please compare this to the [WEL for your gas](#) (EH40). Detection is needed if the calculated value exceeds the WEL.

Common WELs: Ammonia 25 ppm, Carbon Monoxide 20 ppm, Hydrogen Sulphide 5 ppm.

Your risk assessment should also consider local environmental influences, including how the toxic gas disperses, ventilation patterns, and any confined spaces where exposure concentrations could rise.

Consider the location of the cylinder, and how closely a person must work in relation to it. There may be instances where the calculator may not suggest detection is necessary due to the room concentration being below a threshold, however exposure limits for toxic gases are generally low enough such that a user can exceed the WELs within seconds after exposure even to small leaks.

If we go to EH40 (linked in the calculation box within the calculator), we should check the applicable Carbon Monoxide exposure limits listed there. In practice, both the long-term and short-term exposure limits should be considered when interpreting the calculator output.

Substance	CAS number	Workplace exposure limit	
		Long-term exposure limit (8-hr TWA reference period)	
		ppm	mg.m ⁻³
Cadmium sulphide and cadmium sulphide pigments (respirable dust (as Cd))		-	0.03
Caesium hydroxide	21351-79-1	-	2
Calcium carbonate inhalable dust	1317-65-3	-	10
Calcium carbonate respirable		-	4
Calcium cyanamide	156-62-7	-	0.5
Calcium hydroxide	1305-62-0	-	5
			1
Calcium oxide	1305-78-8	-	2
			1
Calcium silicate inhalable dust	1344-95-2	-	10
Calcium silicate respirable		-	4
Captan (ISO)	133-06-2	-	5
Carbon black	1333-86-4	-	3.5
Carbon dioxide	124-38-9	5000	9150
Carbon disulphide	75-15-0	5	15
Carbon monoxide	630-08-0	20	23
		30	35

As a catastrophic release would be far above the relevant exposure limits, and because local concentrations near the source would also exceed the room average, detection and additional controls would be expected.

If we then click the “Generate Report” button in the calculator and scroll down, it can also provide recommendations for different types of detection as well as heights.

In this instance, the detector to be used would be an electrochemical sensor that would detect CO, and it would ideally be mounted in the breathing zone (between 1.5m – 1.8m) as the molecular weight of the gas is similar to that of air.

Gas Detection Guidance

If the recommendations above indicate that fixed gas detection is required, consider the following selection and placement criteria:

DETECTOR TYPE

- **ASPHYXIANTS (N₂, AR, HE):**
O₂ depletion sensors.
- **CO₂:**
Specific NDIR CO₂ sensors (O₂ sensors will not detect dangerous CO₂ levels in time).
- **FLAMMABLES:**
Pellistor (catalytic bead) or Infrared sensors calibrated to the target gas.
- **TOXICS:**
Electrochemical sensors specific to the gas.

MOUNTING HEIGHT

- **HEAVIER THAN AIR (E.G. CO₂, AR, PROPANE):**
Mount low, approx. 300mm from the floor or in pits/trenches.
- **LIGHTER THAN AIR (E.G. HE, H₂):**
Mount high, near the ceiling or exhaust extraction points.
- **SIMILAR TO AIR (E.G. N₂, CO, O₂):**
Mount at typical breathing zone height (approx. 1.5m - 1.8m).

Scenario 7: Slow Leak, Time to Hazard & Heavy Gas Detection

Answer:

- **Steady-State Concentration:** 0.11%.
- **Time to Hazard (to 0.5%):** Approximately 2hrs 18 mins.
- **Detector Type & Placement:** If detection were required, the calculator guidance indicates that a specific NDIR CO₂ sensor mounted low (approx. 300 mm from the floor) would be appropriate.
 - Methodology: Standard oxygen-depletion sensors do not replace dedicated CO₂ sensors. Although CO₂ will eventually mix with room air, it can accumulate at low level when released, particularly when cold, so CO₂-specific detection is required. The onset of the risk to health from CO₂ gas is also much faster than the asphyxiation risk, and therefore it is important that CO₂ leaks are detected as fast as possible.

Explanation:

As previously mentioned, calculating the volume of a slow gas leak is not an exact science, and the calculator should therefore be used with caution. If key values are not known or cannot be measured, it becomes difficult to draw accurate conclusions about the risk to individuals.

Where possible, leak rates and air-change rates should be measured rather than estimated. If this is not reasonably practicable, a conservative worst-case approach should be applied. In this scenario, we know the gas, the size of the leak and the air changes we have (assuming all have been measured/confirmed).

Once again, we use the option for “I want to calculate for a Slow Leak / Constant Release with Ventilation.”

We can enter the free air volume but will need to calculate the leak rate. We can do this by selecting 'Estimate Leak Rate' and entering the regulator gauge pressure and equivalent hole diameter from the scenario.

Leak Scenario

Gas Line (Post-Regulator) ▼

Which calculation should I use? Use **Gas Line (Post-Regulator)** when the line contains gas, for example nitrogen that has already been vaporised and pressure-reduced. Use **Liquid Nitrogen Line** only when the line is carrying **liquid nitrogen** before it flashes to gas at the leak point.

Gas Line (Post-Regulator)

Gas Type

Carbon Dioxide ▼

Pressure (Post-Regulator) (bar)

Use the actual downstream line pressure

2.5 ▴ ▾

Equivalent Hole Diameter (mm)

0.2 ▴ ▾

Upstream Regulator Max Flow (m³/hr)

Optional logic check

Optional ▴ ▾

Note: We enter the regulator gauge pressure reading in bar(g), exactly as shown on the regulator, not a separately adjusted line pressure.

Estimate Leak Rate

Use This Leak Rate

Estimated Leak Rate: 0.04368 m³/hr

We then click Estimate Leak rate, which will be shown underneath the button. You can then click “Use this Leak Rate” to feed the value into the calculator.

We can then enter the Air Changes per Hour, selected Carbon Dioxide (Enrichment), and click Calculate Concentration.

Free Air Volume of room (m³)

20

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)



The **free air volume** is the room volume minus the space occupied by furniture and equipment. The most accurate way to obtain this value, is to directly measure it. There are some phone apps that may help you do this. If unknown, enter the room volume and apply the 40% reduction checkbox as a conservative estimate.

Slow Leak Parameters

Leak Rate (m³/hr)

Small ≈ 0.001 | Medium ≈ 0.1 | Large ≈ 1 | Catastrophic ≈ 100

0.04368

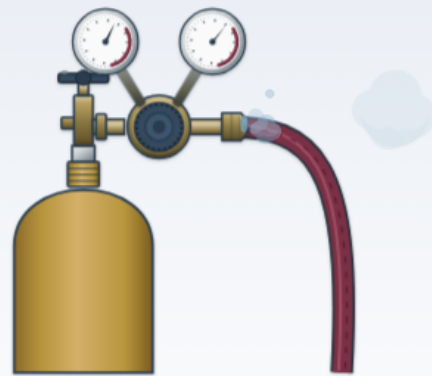
Estimate Leak Rate

Air Changes per Hour (ACH)

2

Note: This calculation assumes perfect mixing of the gas within the room.

If ventilation is poor, uneven, or uncertain, the calculated room concentration may understate the local hazard.



For a **slow leak** (e.g., from a regulator or valve), gas mixes continuously with room air. Enter the **leak rate** and **ACH** to find the steady-state concentration.

Gas Hazard Type

Carbon Dioxide (Enrichment)



 Calculate Concentration

 Generate Report

Gas concentration in room: **0.11%**

● CO₂ detection is not likely to be required. Still, ensure your risk assessment considers local features where CO₂ may settle, including basements, cellars, and enclosed low spaces.

⚠ Worst-Case Time to Hazard (if ventilation fails)

🕒 Time to 0.5% CO₂ (detection): **2 hrs 18 min**

🕒 Time to 4% CO₂ (IDLH): **18 hrs 41 min**

Based on 0 ACH (sealed room). Assumes the leak continues uninterrupted with no automatic shutdown, isolation, or manual intervention.

Screening note: This is a screening estimate to support risk assessment. Average room concentration may not reflect the most severe local exposure near the source, in low/high level spaces, or in poorly ventilated connected voids. Where significant risk remains, prioritise elimination, substitution, segregation, and ventilation before relying on detection.

Personal Bubble Assessment (2×2×3 m)

Gas concentration near leak: **0.18%**

● Localised CO₂ levels near the leak are within acceptable limits. A person working or passing near this source is unlikely to be at significant risk, however CO₂ is heavier than air and will accumulate in low areas — take care if working at floor level near this source.

** Note: The bubble models the localised area (12 m³) near the leak source before full mixing. It uses the room ACH to account for active dilution within this smaller zone, which is why it often shows a higher concentration than the whole room average.*

⚠ Worst-Case Time to Hazard (if ventilation fails)

🕒 Time to 0.5% CO₂ (detection): **1 hr 23 min**

🕒 Time to 4% CO₂ (IDLH): **11 hrs 13 min**

Based on 0 ACH in the 12 m³ personal zone. Assumes the leak continues uninterrupted with no automatic shutdown, isolation, or manual intervention.

The steady-state room concentration is calculated as 0.11%, indicating that the room-average risk after full mixing is relatively low on a screening basis.

In the event of a complete ventilation failure, gas would begin to accumulate in the room, as the ability for contaminated air to be removed is lost. In this scenario, the calculator provides a "Time to Hazard" estimate, indicating that the room would reach the 0.5% CO₂ threshold after approximately 2 hours and 18 minutes. This value represents a warning threshold for further action, not an indication that conditions remain acceptable up to that point.

Review of the generated report also provides guidance on the appropriate type of gas detection, although this scenario does not result in a blanket recommendation for detection. Where

detection is required, the guidance indicates that an NDIR CO₂ sensor should be used and mounted at low level, typically around 300 mm above floor height.

Gas Detection Guidance

If the recommendations above indicate that fixed gas detection is required, consider the following selection and placement criteria:

DETECTOR TYPE

- **ASPHYXIANTS (N₂, AR, HE):**
O₂ depletion sensors.
- **CO₂:**
Specific NDIR CO₂ sensors (O₂ sensors will not detect dangerous CO₂ levels in time).
- **FLAMMABLES:**
Pellistor (catalytic bead) or Infrared sensors calibrated to the target gas.
- **TOXICS:**
Electrochemical sensors specific to the gas.

MOUNTING HEIGHT

- **HEAVIER THAN AIR (E.G. CO₂, AR, PROPANE):**
Mount low, approx. 300mm from the floor or in pits/trenches.
- **LIGHTER THAN AIR (E.G. HE, H₂):**
Mount high, near the ceiling or exhaust extraction points.
- **SIMILAR TO AIR (E.G. N₂, CO, O₂):**
Mount at typical breathing zone height (approx. 1.5m - 1.8m).

Scenario 8: Calculation Risk of Exposure when handling Dry Ice

Answer:

Maximum volume of CO₂ gas generated: 0.108m³

Local Risk Present? The personal bubble assessment indicates a local concentration of 0.9%, so the calculator supports further assessment rather than a blanket conclusion that the task is low risk. The real significance depends on sublimation rate, duration, local airflow, and the worker's breathing-zone position relative to the dry ice.

Explanation:

For this calculation, we will look to use the Dry ice calculator, which can be accessed by clicking the “I have **dry ice (solid CO₂)** and want gas volume from a known mass.” button.

We know that we have 200g of Dry ice, which is 0.2kg – we enter that into the field and click Calculate Volume. The calculator returns a gas volume of 0.108m³.

We can then continue down the calculator to determine the room concentration following equilibration, as well as the personal bubble assessment.

As the free air volume is already known, it is entered into the relevant field. The calculated gas volume is automatically populated, and the Gas Hazard Type is pre-selected.



Calculate Volume - Dry Ice

From solid CO₂ mass

Dry ice mass (kg)

0.2

 Calculate Volume

Volume of CO₂ gas generated = **0.108 m³**

✓ Your calculated value will be pre-populated in Step 4 below.

Note: The UoN Health and Safety Department does not endorse the use of dry ice in beverages.



Calculated using an approximate dry ice expansion of **5.4 m³**
CO₂ per 10 kg (0.54 m³/kg).

Click "Calculate Concentration" to obtain both the room-average and personal bubble concentrations. In this scenario, we are primarily interested in the personal bubble result, which is calculated as 0.9%.

Note: The calculator provides a generalised indication of potential risk, and the specific activity being undertaken must always be considered carefully. The personal bubble assumes a 2m x 2m x 3m zone around the source of release; however, if a user is handling dry ice close to their breathing zone (for example, within 50cm), the actual local risk may be higher than this simplified assumption suggests. In addition, carbon dioxide gas generated from sublimation may initially accumulate at low level rather than rising and dispersing immediately.

This calculation assumes an instantaneous release and therefore represents a worst-case scenario. As the calculated concentration exceeds the 0.5% threshold, the output indicates that further risk assessment is required, taking into account the factors described above.

This calculation is for instantaneous release and calculates a worst-case scenario. Given the limit of 0.5%, we would appear to be above this so this calculation would determine that a further assessment of risk is necessary. For this, we need to expand our analysis to include the factors mentioned above.



Personal Bubble Assessment (2×2×3 m)

Gas concentration near leak: **0.9%**

■ Significant risk to anyone working in close proximity to this leak. Even brief exposure at this level can cause headaches, dizziness, and shortness of breath. Prioritise elimination of the hazard or improving ventilation. Personal CO₂ monitors should only be considered if the work is in a confined space and subject to an approved permit to work.

For larger quantities of dry ice, a more detailed assessment may be required. In such cases, it may be appropriate to consider the sublimation rate to better understand potential exposure, recognising that this introduces additional complexity beyond this screening approach.

Scenario 9: Mixed Gas Cylinders & Multiple Independent Risks

Answer:

Gas Volume: 9.67m^3 . (Using the supplier volume for the cylinder contents.)

CO₂ Hazard: The CO₂ gas volume is 0.967m^3 (10% of total). This results in a room CO₂ concentration of 3.22%, which is well above the calculator's 0.5% CO₂ threshold.

Hydrogen Hazard: The H₂ gas volume is 0.967m^3 (10% of total). This results in a room H₂ concentration of 3.22%, which is above 25% of the LEL for hydrogen (1%).

Oxygen Depletion: The total gas release (9.67m^3) displaces air, resulting in a room O₂ concentration of approximately 14.20%.

Detector Strategy: All three hazards are significant, but they do not reach their respective thresholds at the same point. On a room-average basis, the CO₂ threshold is reached first, followed by oxygen depletion, and then the hydrogen 25% LEL threshold.

Recommendation: On a room-average screening basis, a CO₂ detector is likely to provide the earliest alarm. That does not make oxygen depletion or hydrogen-related controls automatically redundant; the final detector strategy should be confirmed from the actual release scenario, room layout, ventilation pattern, and work location.

Explanation:

When assessing mixed-gas cylinders, it is important to understand both the physical properties of the cylinder and the hazards presented by each gas component.

In this scenario, the cylinder is a standard BOC size 'L' cylinder. BOC provides this value directly for this gas mixture, stating that the cylinder contains approximately 9.67m^3 of gas in atmospheric conditions.

Although a slightly different value (e.g. 9.87m^3) could be calculated using the pressure of the cylinder and the water capacity, the supplier-stated free air volume is preferred, as it accounts for real-gas behaviour and filling practice. For this reason, 9.67m^3 is used in the calculation.

The gas mixture comprises 10% CO₂, 10% H₂ and 80% N₂ so the individual component volumes are:

1. CO₂ - 0.967m^3
2. H₂ - 0.967m^3
3. N₂ - 7.74m^3

As all components act as asphyxiants, the oxygen depletion risk is assessed using the total released volume (9.67m^3).

Calculate Concentration in Room

Enter room volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

☐ Apply 40% reduction for conservative estimate (if free air volume unknown)

Volume of gas (m³)

Gas Hazard Type

The **free air volume** is the room volume minus the space occupied by furniture and equipment. If unknown, apply the 40% reduction checkbox as a conservative estimate.

This results in a calculated room oxygen concentration of approximately 14.2%, indicating a risk of asphyxiation.

O₂ concentration in room: **14.20%**

● O₂ detection necessary. Consider local factors such as room ventilation, enclosure volume, and whether the displaced oxygen might stratify or accumulate based on the gas used.

The CO₂ enrichment risk is then assessed separately using the CO₂ component volume (0.967m³), resulting in a room concentration of 3.22%, which exceeds the calculator's 0.5% CO₂ threshold.

O₂ concentration near leak: **4.07%**

● Immediately dangerous for anyone near this leak. Oxygen levels in the personal zone would drop to a level that causes loss of consciousness within seconds and can be fatal. No person should approach, work near, or pass through this area without a suitably planned and rehearsed confined space action plan.

Similarly, this indicates we have the potential to create a hazardous environment in the event of a catastrophic release.

Finally, the hydrogen flammability risk is assessed using the hydrogen component volume (0.967m^3). This produces a room concentration of 3.22%, exceeding 25% of the Lower Explosive Limit (1%).

Gas concentration in room: **3.22%**

 Please compare this to the [LEL for your gas](#) to confirm whether detection is required. Detection is required if the calculated percentage exceeds 25% of the LEL.

Common LELs: Acetylene 2%, Hydrogen 4%, Methane 5%.

Your risk assessment should also consider local conditions relevant to your specific flammable gas, such as whether it tends to rise or sink, how ventilation affects dispersion, and the location of potential ignition sources.

In this instance, we are at 3.22%, which exceeds 25% of the LEL for H_2 (1%).

As we can see, all three identified hazards have the potential to create a safety risk, but for different reasons.

In practice, a mixed-gas release should not be thought of as each component immediately behaving in isolation. The escaped gas will initially behave as a plume, and its movement will depend mainly on the overall properties of the released mixture, the release momentum, ventilation, and the room layout.

The calculator is therefore most useful for identifying which hazard threshold is reached first on a screening basis. That helps indicate which detector may provide the earliest warning, but it does not by itself prove that one detector is always sufficient or that other hazards can be ignored.

On a room-average screening basis, the CO_2 threshold is reached first, indicating that a CO_2 detector is likely to provide the earliest general warning. However, this does not mean that oxygen-depletion or hydrogen-related controls are redundant. The final detector strategy should be determined by the specific release scenario, room layout, ventilation pattern, and work location and should be confirmed by a competent gas detection specialist.

Scenario 10: Dense Gas Stratification

Answer:

1. 0.33m
2. Room Vol = 45m^3 ; Free Air Volume = 27m^3
3. O_2 concentrations:
 - a. 0% within volume of room no below 0.33m gas cloud height.
 - b. 18.62% after equilibration in room.
4. It wouldn't appear at though there is any risk, but in reality, the gas will rise over time, and the concentration a worker would be exposed to would be somewhere between 0% and 18.62% as time passes, so they would likely be in a deficient atmosphere.

Explanation:

In this scenario, we will make use of the Dense gas calculator to predict the risk of working in environments where there is potential for dense gases to collect.

1. For dense gases, they will form a cloud of a specific minimum height based on the area the gas will accumulate in, and the volume of the gas released.

The area of the room floor is 15m^2 , but this is not the free air area. We are going to assume a 40% reduction of volume is needed to take into account the space taken up by equipment, and therefore we can reduce 15m^2 by 40%, so give us a floor area of 9m^2 . It is worth highlighting, once again, that measurements are always preferable and will be more accurate.

The volume of the gas in the cylinder is 3m^3 , so the gas cloud height can be calculated by dividing 3m^3 by 9m^2 , and therefore the initial stratified gas layer height is 0.33m.

Note: In reality, the cloud will evolve over time and disperse itself uniformly throughout the room as a result of passive diffusion and other external influences like ventilation. We will need to use both the Dense gas calculation as well as the room concentration, together, to determine the real risk to a person working in this lower area.

2. The volume can be calculated by multiplying the area of the room by the ceiling height, so 15m^2 multiplied by 3m, which equals a room volume of 45m^3 .
We will use the 40% reduction built into the calculator, so we can enter 45 into the field and tick the 40% reduction (which will equal a free air volume of 27m^3).
3. We now have the information we need to carry out the calculation. We know the volume of the contents, so we can select the option in the calculator corresponding to this.

We can enter 45m^3 as the volume of the room and then tick the 40% reduction box to obtain a conservative estimate of the free air volume.

Next, we check the Dense Gas option – this calculation sets a false ceiling height so allows the user to calculate the concentration of the gas (or remaining oxygen) within the volume related to this lower ceiling height. We know from the scenario that the ceiling height is 3m, and we calculated in part 1 that we're expecting a minimum gas stratification height of 0.33m.

We can then enter the Volume of gas into the final field.



Calculate Concentration in Room

Enter free air volume, gas hazard type, then calculate

Enter the volume of the room and select the gas type, then click Calculate.

Free Air Volume of room (m³)

45

☒ Apply 40% reduction for conservative estimate (if free air volume unknown)

☒ Dense Gas (advanced) 

Ceiling Height (m)

3

Expected Gas Stratification height (m)

0.33

Optional. Defaults to 1.0 m. Enter the height at which you expect gas to stratify. The calculator will estimate the concentration within the room volume below this height. Use a lower value where gas may collect close to the floor, such as in pits, trenches, floor-level work, or very cold/heavy gas releases.

Volume of gas (m³)

3

We select Asphyxiant, as we are using Argon gas which displaces oxygen, and then click Calculate Concentration.

Room-average basis for detection recommendation: O₂ concentration in room: **18.62%**

 Further assessment of risk necessary. Your assessment should account for local conditions including airflow, potential gas build-up, and how the specific asphyxiant behaves in the space. Consider the quality of ventilation and whether there are any dead zones where gas may collect, as well as whether smaller cylinders can be used.

 **Dense Gas (advanced)**

EXPECTED GAS STRATIFICATION

Concentration estimate

O₂: 0.00%

Stratification height: 0.33 m.

We can see from the calculation that if the gas were to perfectly stratify at a height of 0.33m, then the O₂ concentration within that area will become 0% and therefore be a fully deficient area which would not be breathable.

If the gas were to fully equilibrate throughout the room, it would reach 18.62%, which would be below the legal limit and require further investigation.

The calculator suggests that the gas stratifies perfectly into a cloud of height 0.33m, and within that height the atmosphere contains 0% oxygen. Under perfect conditions, this is likely to be true, but gas stratification is a transient effect. Depending on a number of factors, namely any ventilation present, any air flow from movement of people/equipment, temperature, etc, the gas is likely to diffuse out of a perfect, stratified cloud and distribute more uniformly as time passes.

Keeping in line with the assumptions the calculator makes, namely no ventilation and therefore the gas does not escape the room, a realistic scenario is that a person working in the lower tier of this room could temporarily be working in a breathing zone that contains 21% oxygen, but as the gas begins to equilibrate this could drop quite substantially to anywhere between 0% and 18.62%, which are the two values calculated.

Therefore, there is a likelihood that any person working at this height could be at substantial risk of asphyxiation under certain circumstances. Depending on the frequency and nature of this job, consideration should be given to use of personal detection and a confined space risk assessment and permit to work may be required if work is necessary.

Part 3: Gas/Cryogen Safety Calculator: Methodology, Equations and Assumptions

This document outlines the theoretical models and formulas utilised by the Gas/Cryogen Safety Calculator, based on the methodology detailed in example questions.

1. Volume Calculations

Before any concentration can be determined, the total volume of gas (measured at standard atmospheric pressure and temperature) must be calculated from the initial source parameters.

1.1 Compressed Gas Volume

This calculation is approximated using Boyle's Law ($P_1V_1 = P_2V_2$), assuming constant temperature, expanding the cylinder's internal capacity to standard atmospheric pressure.

$$V_{gas} = \frac{P_{cyl} \times \left(\frac{V_{water}}{1000}\right)}{1.013}$$

Where:

V_{gas} = Total expanded volume of the gas (m^3)

P_{cyl} = Initial pressure of the cylinder (bar)

V_{water} = Water capacity of the cylinder (Litres)

1000 = Conversion factor (Litres to m^3)

1.013 = Standard atmospheric pressure (bar)

1.2 Liquefied Gas Volume

A direct mass-to-volume conversion using the specific volume of the gas at standard conditions.

$$V_{gas} = W \times SV$$

Where:

V_{gas} = Total expanded volume of the gas (m^3)

W = Weight of the liquefied contents (kg)

SV = Specific Volume of the specific gas (m^3/kg)

1.3 Cryogenic Gas Volume

Utilises standard liquid-to-gas expansion ratios.

$$V_{gas} = \frac{V_{liquid} \times E_R}{1000}$$

Where:

V_{gas} = Total expanded volume of the gas (m^3)

V_{liquid} = Volume of the cryogenic liquid (Litres)

E_R = Expansion Ratio (e.g., 694 for Liquid Nitrogen, 757 for Liquid Helium, 535 for Liquid CO₂)

1000 = Conversion factor (Litres to m³)

2. Room Concentration (Instantaneous Release)

This model assumes a catastrophic failure where the entire container contents are released immediately and mix perfectly into the room.

2.1 General / Flammable / CO₂ Concentration (%)

$$C_{\%} = \left(\frac{V_{gas}}{V_{room}} \right) \times 100$$

Where:

$C_{\%}$ = Resulting concentration of the gas in the room (%)

V_{gas} = Total expanded volume of the gas (m³)

V_{room} = Free air volume of the room (m³)

2.2 Toxic Gas Concentration (ppm)

$$C_{ppm} = \left(\frac{V_{gas}}{V_{room}} \right) \times 1,000,000$$

Where:

C_{ppm} = Resulting concentration of the gas in the room (parts per million)

2.3 Oxygen Deficiency (Asphyxiant Concentration)

This calculation assumes that the gas released from the cylinder forces an equivalent volume of perfectly mixed 'normal air' out of the room. It assumes normal clean air contains exactly 20.95% oxygen by volume in an instantaneous release.

$$O_{2(\%)} = \left(\frac{(V_{room} - V_{gas}) \times 0.2095}{V_{room}} \right) \times 100$$

Where:

$O_{2(\%)}$ = Resulting oxygen concentration in the room (%)

0.2095 = Standard volume fraction of oxygen in normal air (20.95%)

2.4 Oxygen Enrichment

This calculates the concentration when pure oxygen is released into a room, accounting for the displacement of standard room air (assumed 20.95% baseline for this specific calculation).

$$O_{2\%} = \left(\frac{(V_{room} \times 0.2095) + V_{gas} - (0.2095 \times V_{gas})}{V_{room}} \right) \times 100$$

3. Continuous Slow Leak (Steady-State)

This models a steady leak into a room with active ventilation. Based on a simplified steady-state mass balance equation, the room reaches 'steady state' when the rate of the hazardous gas entering exactly equals the rate of mixed air extracted.

3.1 Ventilation Rate (Q)

$$Q = V_{room} \times ACH$$

Where:

Q = Volumetric flow rate of ventilation (m³/hr)

ACH = Air Changes Per Hour (hr⁻¹)

3.2 General / Flammable / CO₂ Concentration (%)

$$C_{\%} = \left(\frac{L}{Q} \right) \times 100$$

Where:

L = Leak Rate of the gas into the room (m³/hr)

3.3 Toxic Gas Concentration (ppm)

$$C_{ppm} = \left(\frac{L}{Q} \right) \times 1,000,000$$

3.4 Oxygen Deficiency (Asphyxiants)

Calculates the depressed equilibrium of oxygen levels (using a 20.95% baseline).

$$O_{2\%} = 20.95 \times \left(1 - \frac{L}{Q} \right)$$

3.5 Oxygen Enrichment

Calculates the increased equilibrium oxygen concentration using a 20.95% ambient air baseline and a steady-state mass-balance approach. In the current calculator, oxygen enrichment is treated as an addition above normal air, rather than assuming the incoming oxygen replaces air with 100% oxygen.

$$O_{2\%} = 20.95 + \left(\frac{L}{Q} \times 100 \right)$$

4. Worst-Case Time to Hazard (0 ACH)

This calculates the time required to reach a dangerous threshold if all building ventilation completely fails (0 ACH worst-case). It should be treated as a screening model with no ventilation removal, not as a literal sealed or pressurised box.

4.1 Oxygen Deficiency (Time to 19.5% and 18%)

Utilises a logarithmic decay function to find when the oxygen level drops to specific warning and danger thresholds.

Time to 19.5% (Warning Level):

$$t_{19.5} = \left(\frac{V_{room}}{L} \right) \times \ln \left(\frac{20.95}{19.5} \right)$$

Time to 18.0% (Danger Level):

$$t_{18.0} = \left(\frac{V_{room}}{L} \right) \times \ln \left(\frac{20.95}{18.0} \right)$$

Where:

t = Time required to reach the target concentration (hours)

ln = Natural logarithm

4.2 Gas Concentration (Time to 0.5% and 4% CO₂)

Utilises a logarithmic accumulation function for gases displacing room air.

Time to 0.5% (Detection/Warning):

$$t_{0.5} = - \left(\frac{V_{room}}{L} \right) \times \ln(1 - 0.005)$$

Time to 4.0% (Immediately Dangerous to Life or Health (IDLH) limit):

$$t_{4.0} = - \left(\frac{V_{room}}{L} \right) \times \ln(1 - 0.04)$$

4.3 Oxygen Enrichment (Time to 23.5%)

Calculates the time until standard air is enriched to the critical 23.5% threshold.

$$t_{23.5} = - \left(\frac{V_{room}}{L} \right) \times \ln \left(\frac{100 - 23.5}{100 - 20.95} \right)$$

4.4 Flammable / Toxic Linear Accumulation

Because standard exposure limits (LEL and WEL) vary drastically by specific gas, the calculator outputs a linear accumulation rate per hour for the user to compare against their specific gas limits.

Flammable Rate (% per hour):

$$Rate_{\%} = \left(\frac{L}{V_{room}} \right) \times 100$$

Toxic Rate (ppm per hour):

$$Rate_{ppm} = \left(\frac{L}{V_{room}} \right) \times 1,000,000$$

5. Dry Ice Conversion

The dry ice calculator converts solid CO₂ mass to equivalent gas volume using a fixed expansion factor of 0.54 m³/kg. This is intended as a practical approximation for the maximum gas volume generated by sublimation.

Where:

$$V_{gas} = m_{dry} \times 0.54$$

m_{dry ice} = mass of dry ice (kg)

6. Advanced Leak-Rate Estimator

The advanced leak-rate estimator provides a rough leak-rate input for the Slow Leak calculation when a measured leak rate is not available. It is a screening method only and should not be treated as a detailed flow-calculation tool.

6.1 Gas Line (Post-Regulator)

For choked flow (gas escaping through small hole & reaches the speed of sound) post-regulator gas leaks, the calculator estimates leak rate from regulator gauge pressure, equivalent hole diameter, and a gas factor. The gauge pressure is first converted internally to absolute pressure.

$$Q = 0.4 \times D^2 \times (P_{gauge} + 1) \times F_g$$

Where:

P_{gauge} = Regulator gauge pressure in bar(g)

D = Equivalent hole diameter (mm)

F_g = Gas-specific factor

This calculator uses an approximation formula for quick estimates. It assumes choked flow through a standard sharp-edged orifice at room temperature (~20°C). The baseline calculation is optimised for the thermodynamic properties of Air, but when selecting other gases, the factor adjusts for molecular weight.

Because of the approximations used, there may be an error margin of up to, approximately, 6% depending on the gas.

6.2a Liquid Nitrogen Line

For LN₂ line leaks, the calculator estimates the equivalent gaseous nitrogen hazard rate for oxygen-deficiency screening rather than the liquid flow rate itself.

Where:

$$Q_{LN_2, gas} = 19.3 \times D^2 \times \sqrt{P_{gauge}}$$

P_{gauge} = Gauge pressure in bar

D = Equivalent hole diameter (mm)

6.2b Liquid CO₂ Line

For LCO₂ line leaks, the calculator estimates the equivalent gaseous carbon dioxide hazard rate for CO₂ enrichment screening rather than the liquid flow rate itself.

Where:

$$Q_{LCO_2,gas} = 14.1 \times D^2 \times \sqrt{P_{gauge}}$$

P_{gauge} = Gauge pressure in bar

D = Equivalent hole diameter (mm)

6.3 Regulator Maximum Flow Cap

If an upstream regulator maximum flow is entered, the calculator applies a logic cap so that the leak rate used in the Slow Leak model cannot exceed the maximum flow the regulator can physically supply.

7. Recommendation and Threshold Logic

The calculator does not use a single generic risk rule for every gas. It applies different screening thresholds depending on the selected hazard type.

Asphyxiants: room oxygen below 19.5% prompts further assessment, and below 18% indicates a severe oxygen-deficiency risk.

Oxygen enrichment: oxygen above 23.5% indicates an enrichment risk requiring control.

CO₂: room concentration above 0.5% triggers the calculator CO₂ warning threshold, with higher local concentrations indicating more serious proximity risk.

Flammables: the calculator reports concentration and asks the user to compare it with the gas-specific LEL, including 25% of the LEL as the key screening threshold used in the current interface.

Toxics: the calculator reports ppm and requires comparison against the applicable occupational exposure limits for the specific gas.

8. Personal Bubble Assessment

A secondary calculation applied simultaneously to all scenarios to estimate the localised concentration immediately surrounding the leak.

It uses the exact same equations outlined in Sections 2, 3 and 4, but applies them to a smaller fixed zone.

The room volume (V_{room}) is strictly replaced with 12m³ (representing an arbitrary 12 m³ workspace / 2 x 2 x 3 m zone).